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ORIGINAL INVESTIGATION

Estimating Muscle Fiber-Type Composition in Elite Athletes: A Survey on Current Practices and Perceived Merit

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Purpose: To gather information on practices and perceptions of high-performance experts regarding their athletes' muscle fiber-type composition (MFTC) and its estimation. ***Methods:*** A questionnaire on the noninvasive versus invasive estimation of MFTC was completed by 446 experts including coaches and sport-science/sports-medicine staff. Moreover, the perceived importance of MFTC for training and performance optimization was assessed. Differences between sport types (individual and team sports) were analyzed using chi-square tests. ***Results:*** Forty percent of the experts implemented MFTC assessment in pursuit of performance optimization, while 50% did not know their athletes' MFTC but expressed a desire to implement it if they would be able to assess MFTC. Ten percent did not perceive value in MFTC assessment. Only 18% of experts believed that their athletes would undergo a muscle biopsy, leading to the adoption of alternative noninvasive techniques. Experts primarily relied on their experience to estimate MFTC (65%), with experts working in individual sports using their experience more frequently than those working in team sports (68% vs 51%; $P = .009$). Jump tests emerged as the second-most commonly employed method for estimating MFTC (56%). When only considering experts who are currently using MFTC, 87% use MFTC to individualize training volume and 84% to individualize training intensity. ***Conclusions:*** Experts value MFTC assessment primarily to individualize training but mainly rely on noninvasive methods to estimate MFTC. Some of these methods lack scientific validity, suggesting a continuing need for education and further research in this area.

Keywords: individual sport, team sport, muscle biopsy, noninvasive

Human skeletal muscle consists of a mosaic of slow- and fast-twitch fibers, which have distinct characteristics. Fast-twitch fibers (type II) contain myosin heavy-chain IIa (MyHC-IIa) or MyHC-IIx, are predominantly glycolytic, and have a high contraction velocity and power, while slow-twitch fibers (type I) contain predominantly MyHC-I, are mostly oxidative and energy efficient, and have a high fatigue resistance.[1] The average human vastus lateralis muscle comprises approximately 50% slow- and 50% fast-twitch fibers. However, a large heterogeneity exists, resulting in some athletes having a majority of either slow- or fast-twitch muscle fibers.[2] This distribution is largely determined by genetics and can only be moderately influenced by training.[3]

Since the classical studies in the 1970s, human muscle fiber type composition (MFTC) is known to be an important sport performance determinant.[4] MFTC was found to be different for athletes specializing in either sprint or endurance disciplines, as well as between different sports.[5–8] In that period, muscle biopsies were briefly used in routine sport practice by a limited number of countries for talent identification and training monitoring.[9] However, for obvious reasons, this invasive muscle fiber typing via biopsies in athletes has largely been discontinued, except in an academic research settings.[10] In subsequent decades, however, several noninvasive estimators have become available through technological as well as data-driven advancements that could contain accurate information on MFTC in athletes.[11–17] For example, single[18,19] and repeated jump tests[12] and proton magnetic resonance spectroscopy of carnosine[11] have been shown to be good estimators of MFTC. Using the latter technique, new links between the MFTC and race tactical decisions, recovery optimization, and muscle injuries were discovered, emphasizing its importance in sports performance.[20–25]

Nevertheless, it is unclear whether coaches and sport science/sport medicine (SSSM) staff see merit in knowing MFTC of their athletes. Moreover, little is currently known about how experts try to measure/estimate MFTC of their athletes. Therefore, the aim of the study was to survey (1) the current noninvasive versus invasive methods and metrics to estimate MFTC and (2) the beliefs of coaches and SSSM staff on the use and the importance of MFTC in sports practice. Gaining knowledge of the beliefs and common practices to estimate and/or assess MFTC by experts and its application for performance optimization will serve as a basis for future research and education.

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Methods

Subjects

An online questionnaire was sent to coaches and SSSM staff (further referred to as experts) of all sport types through email and social media. In total, 740 participants started the questionnaire with 32 being excluded as being athletes and 262 being excluded as only responding to the first questions on participant characteristics, with final analysis and presented data on 446 respondents. Of all experts, 290 worked in individual sports (individual), 104 in team sports (team), and 52 coached both in individual and team sports (multiple). All participants provided informed consent, and the study was approved by the local ethical committee (Ghent University Hospital; B6702020000814).

Study Design

A cross-sectional survey was used to examine the experts' beliefs and practices pertaining to MFTC. The survey (Supplementary Material [available online]) was designed using an online platform (Survey Monkey, www.surveymonkey.com) in 4 languages to enable wide participation (English, French, Spanish, and Dutch). Questionnaire content validity was developed based on domain expertise of all authors as well as in consultation with coaches. A pilot survey was used to test questions before the release of the final version, and data were collected between February 1 and April 27, 2021. The survey included 5 domains:

1. ***Participant demographics***: This part included educational degree, years of coaching, role (sport-specific coach, part of SSSM staff, both), sports in which they were involved, and the performance and/or fitness level of their athletes (as defined by McKay et al[26]).
2. ***Current implementation of MFTC and perception on biopsy***: Experts were asked whether they used their knowledge about MFTC during training/competitions. Moreover, the experts' opinion on biopsies as well as the perceived willingness of their athletes to undergo muscle biopsies (5-point Likert scale) were assessed. Additionally, the ability of ranking their athletes on a muscle fiber-type continuum from fast to slow (5-point Likert scale) was explored (for more details, see Supplementary Material [available online]).
3. ***General methods to estimate or assess MFTC***: Experts were asked about their use of general categories of methods to estimate MFTC (a full list of all inquired methods in Table 1). Jump and strength tests were investigated in more detail to know the exact used protocols and to understand which specific variables were interpreted to estimate MFTC (eg, for jump tests: maximal jump height, rate of force development, etc).
4. ***Sport-specific methods***: Experts were further divided based on their sport: cycling, rowing, running, swimming, team sports, or other/multiple sports, with answers being analyzed separately for individual, team, or multiple sports. Parameters were subdivided into physiological, training, performance, and biomechanical measures (Table 1). Performance measures were specific for individual sports or team sports. Biomechanical measures were investigated only in individual sports.
5. ***Relevance and application of MFTC***: Experts who were using MFTC in their coaching practice were asked what they were using this information for. All experts were asked how important they thought the influence of the MFTC was in certain decisions (5-point Likert scale).

Statistical Analysis

Chi-square tests were used to analyze whether the following qualitative variables were different among sport groups: gender, level of experts' athletes, educational degree, coaching role, current implementation of MFTC, used methods to estimate the MFTC, and the purposes for which MFTC was used in the context of performance optimization.

Analysis of variance with Tukey post hoc test (equal variances assumed) or Kruskal–Wallis test (equal variances not assumed) was used to investigate whether the following quantitative variables were different among sport groups: years of coaching experience, opinion on biopsies, ranking confidence, and perceived importance of MFTC in the context of performance optimization.

Wilcoxon signed-rank tests were used to investigate which tests were mostly used by experts and for which purposes experts mostly used MFTC.

Paired sample *t* tests were used to investigate whether the opinion of the coach differed from perceived willingness of their athletes to undergo a muscle biopsy and for which purposes experts perceived MFTC the most important.

Because an uneven distribution was observed in the role of the team and the level of the experts' athletes among sport groups, it was investigated for all of the above parameters, whether differences were present depending on the role in the team and the level of the experts' athletes in individual and team sports separately, using the same respective tests.

All statistical analyses were performed using SPSS Statistics (version 29.0). Significance was assumed at $P \leq .05$.

Results

Participant Details

A total of 446 experts were included in the analysis (88.6% male). Experts had 14.8 (10.3) years of coaching experience with 51.7% coaching either world-class athletes (competing in World Championships/Olympic Games) or elite athletes.[26] Sixty-six percent of the experts possessed a university degree, and 14.8% completed a Doctor of Philosophy. Differences between "individual," "team," and "multiple" experts are shown in Table 2.

Experts were further classified based on their coaching role, being either a sport-specific coach (52.1%), part of the SSSM staff (19.6%), or a combination of both (28.3%). Experts who were part of SSSM staff were mostly involved as strength and conditioning coach (40.6%), sport scientist (20.3%), physiologist (13.8%), physiotherapist (10.6%), doctor (4.6%), biomechanist (3.7%), or fulfilled another role in the team (6.5%).

Current Implementation of MFTC and Perception on Biopsy

Forty percent of the experts considered the measured or assumed MFTC of their athletes during training/competition, for example, in talent identification, prescribing training modalities, and pacing. In contrast, 10.1% of all experts did not see value in knowing the MFTC of their athletes. The remaining 50.2% of all experts reported they did not consider MFTC in their current coaching practice. However, if the latter group knew the MFTC of their athletes, they would use this information during training/competitions (Figure 1). No differences were present in team or individual sport experts, nor in the experts' roles or level of the experts' athletes.

Concerning the perception of experts on letting their athletes undergo a muscle biopsy, about one-third either disagreed or strongly

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Table 1 Inquiry on Current Practices for Estimating Muscle Fiber-Type Composition

	General methods	Sport-specific methods	
Direct assessment	Muscle biopsy		
Indirect estimation	Experience/observation during training and competition	**Physiological measures**	
	Visual inspection (ie, body composition)	*All sports*	
	Jump tests	Maximal or mean power/speed on a short test	VO_2max
	Countermovement jump	Time to complete a test/workout	Time to exhaustion
	Squat jump	Aerobic threshold during a test	Maximal power/speed at VO_2max
	Repeated jump sequence	Anaerobic threshold during a test	Muscle oxygenation (eg, NIRS and muscle-oxygen monitor)
	Drop jump	Maximal lactate during a test	
	Standing broad jump	**Training measures**	
	Lateral jump	*All sports*	
	Strength tests	Adaptations to training	Sensitivity to injury
	Functional strength	Speed of recovery within a training session	Fatigue indices during a specified test/workout/game
	Isometric strength	Speed of recovery from an acute training session	
	Isokinetic strength	**Performance measures**	
	^{1}H-MRS	*Individual sport*	*Team sport*
	^{31}P-MRS	Performance in distance vs sprint events	Total distance covered in high accelerations during a game
	Electrostimulation	Maximal acceleration during a short distance	Total distance covered in high decelerations during a game
	Mechanomyography	Reaction time	Total running distance during a game
	Tensiomyography	Pacing strategy	Total sprint distance covered during a game
	Genes		Velocity of the ball
			Force produced while hitting/throwing/kicking a ball
		Biomechanical measures	
		Individual sport	
		Force–velocity profile	
		Maximal cadence or frequency (revolutions/min)	
		Ground contact time	
		Optimal cadence or frequency (revolutions/min)	
		Distance per stride/stroke	

Abbreviations: ^{1}H-MRS, proton magnetic resonance spectroscopy; ^{31}P-MRS, phosphor magnetic resonance spectroscopy; NIRS, near-infrared spectroscopy. Note: Experts were asked to answer yes or no to using the mentioned methods.

Table 2 Participant Details

	Total (N = 446)	Individual (n = 290)	Team (n = 104)	Multiple (n = 52)
Age, y	42.7 (11.7)	43.4 (11.4)[a]	39.9 (11.0)[b]	44.7 (13.9)[a]
Gender (male/female/prefer not to identify)	395/50/1	259/30/1[a]	91/13/0[a]	45/7/0[a]
Experts' experience, y	14.8 (10.3)	14.7 (10.1)[a]	14.1 (9.6)[a]	17.1 (12.9)[a]
Level of experts' athletes, %		[a]	[b]	[a]
World class	38.2	44.7	16.8	44.2
Elite	18.9	16.0	30.7	11.5
National	25.3	24.1	31.7	19.2
Trained	17.7	15.2	20.8	25.0
Highest degree, %		[a]	[a]	[a]
PhD	14.8	14.9	13.5	17.3
University	66.1	64.4	69.2	69.2
Nonuniversity higher degree	13.5	14.2	13.5	9.62
Secondary education	5.17	6.23	3.85	1.92
Primary education	0.45	0.35	0.00	1.92
Experts' role, %		[a]	[b]	[b]
Sport-specific coach	52.1	60.6	38.5	32.7
Part of sports-science/-medicine staff	19.6	15.2	26.9	28.8
Both	28.3	24.2	34.6	38.5

Note: Values are mean (SD) or percentages. Identical superscript letters indicate that there is no significant difference between groups ($P > .05$).

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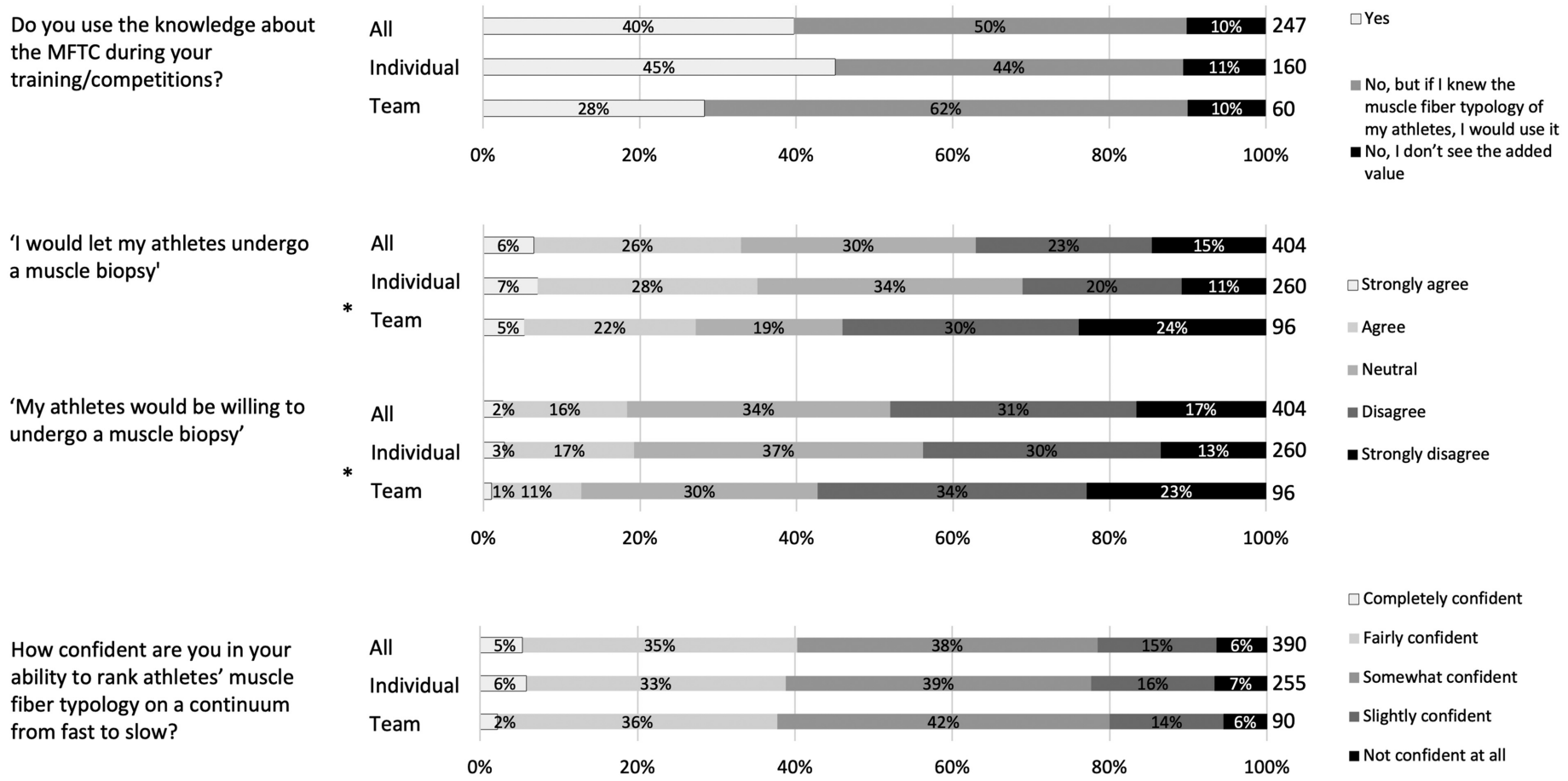

Figure 1 — Experts' use of MFTC, opinion on biopsies, and their confidence in ranking their athletes based on MFTC. The number of experts who answered each question can be seen on the right side of the stacked bar charts. *Significant differences between individual- and team-sport experts. MFTC indicates muscle fiber-type composition.

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disagreed, one-third were neutral, and one-third either agreed or strongly agreed. Team sport experts were less likely to let their athletes undergo a muscle biopsy ($P = .001$) and indicated that their athletes would be less willing to have a muscle biopsy when compared to individual sport experts ($P = .028$). In individual sports, sport-specific coaches thought their athletes would be more willing to undergo a muscle biopsy when compared to the SSSM staff ($P = .009$). Also, experts of elite individual athletes thought their athletes would be more willing to undergo a muscle biopsy when compared to experts training either trained ($P = .006$) or world-class athletes ($P = .003$).

Despite the observed reluctancy on the use of biopsies, 40% of the experts were completely or fairly confident in ranking their athletes' MFTC on a continuum from fast to slow. Team sport experts working with elite athletes were more confident in ranking their athletes than those working with trained athletes ($P = .012$).

General Methods Used to Estimate MFTC

The majority of all experts used their experience/observation during training and competition to estimate MFTC of their athletes (Figure 2). Individual sport experts were more inclined to use their experience (individual: 67.9% vs team: 50.7%; $P = .009$), while team sport experts relied more on jump tests (team: 67.8% vs individual: 47.7%: $P = .001$). In individual sport experts, sport-specific coaches use visual inspection (27.0%) more often than SSSM staff (0.0%) ($P < .001$).

When asking for further details about their jump-based methods (Figure 3A–3C), countermovement jump was most frequently used. Moreover, the maximal jump height (39.5%) was the specific parameter, which was most frequently evaluated by the experts to estimate MFTC. Maximal jump height and reactive strength index were more often used by team sport experts than by individual sport experts (respectively, 47.0% vs 32.2%; $P = .003$ and 26.2% vs 12.9%; $P = .017$). No differences were present when comparing sport-specific coaches versus SSSM staff, nor different levels of the experts' athletes.

When strength exercises were further evaluated (Figure 3B–3D), functional strength exercises were used by the vast majority, which was most evident in individual sport experts when compared to team sport experts (84.3% vs 61.5%, $P = .045$). Of all functional strength exercises that were used, squat, bench press, deadlift, and bench pull were most frequently used. Aside from functional tests, isometric tests (46.8%) were used more often than isokinetic tests (25.5%) to estimate MFTC. The specific parameters of these strength tests to estimate MFTC are very diverse (see Figure 3D). Again no differences were present when comparing different experts' roles, nor the level of the experts' athletes.

Sport-Specific Methods Used to Estimate MFTC

Next to the general methods, we further explored the use of sport-specific methods to estimate MFTC (Figure 4). First, multiple physiological measures were surveyed in which the maximal or

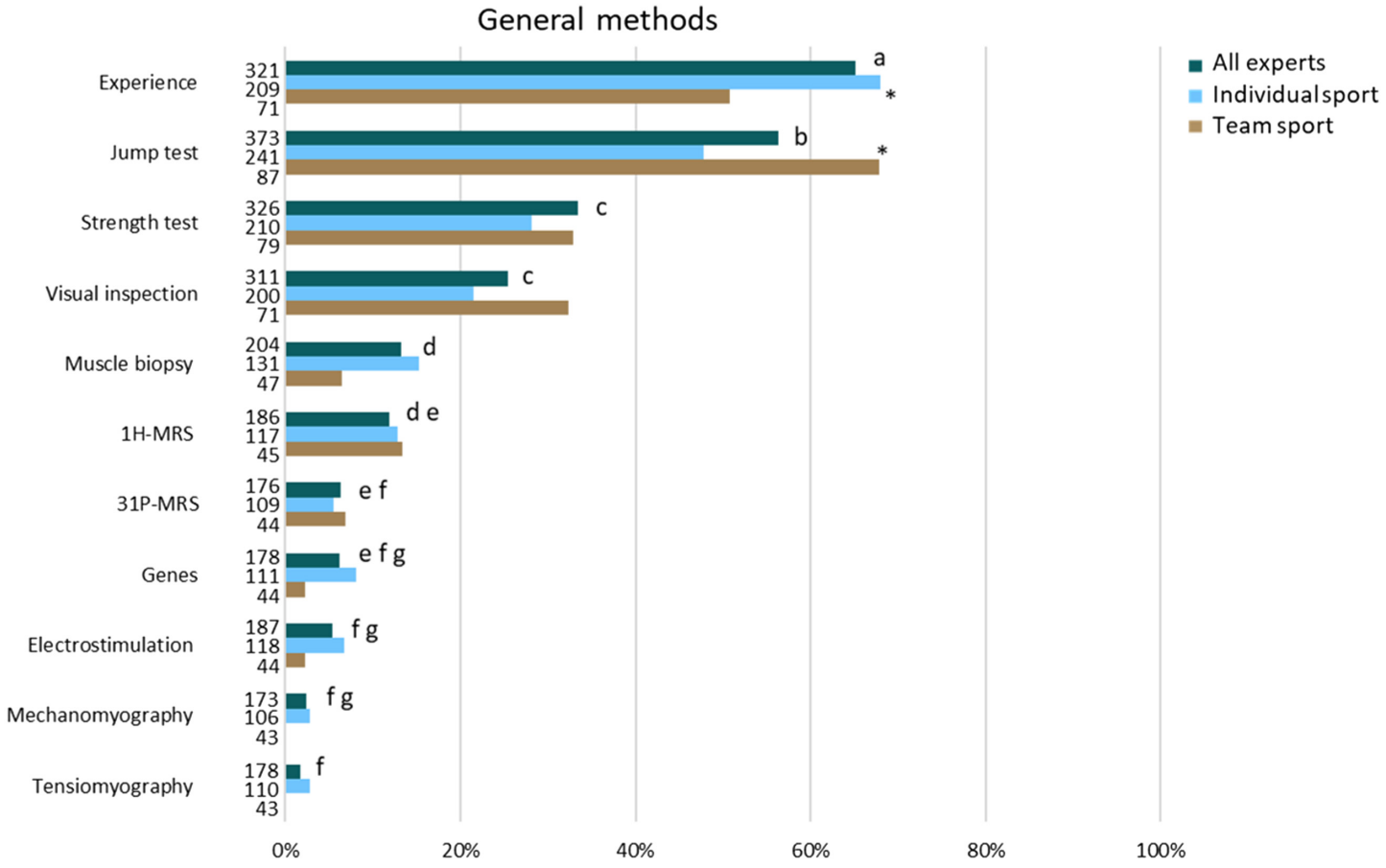

Figure 2 — Experts' use of general methods to estimate the muscle fiber-type composition in their athletes. Number of experts who answered the question can be seen on the left side of the bar chart. *Significant differences between individual and team sport experts. Identical letters indicate that there is no significant difference in the percentual use of different methods in the total group of experts. ^{1}H-MRS indicates proton magnetic resonance spectroscopy; ^{31}P-MRS, phosphor magnetic resonance spectroscopy.

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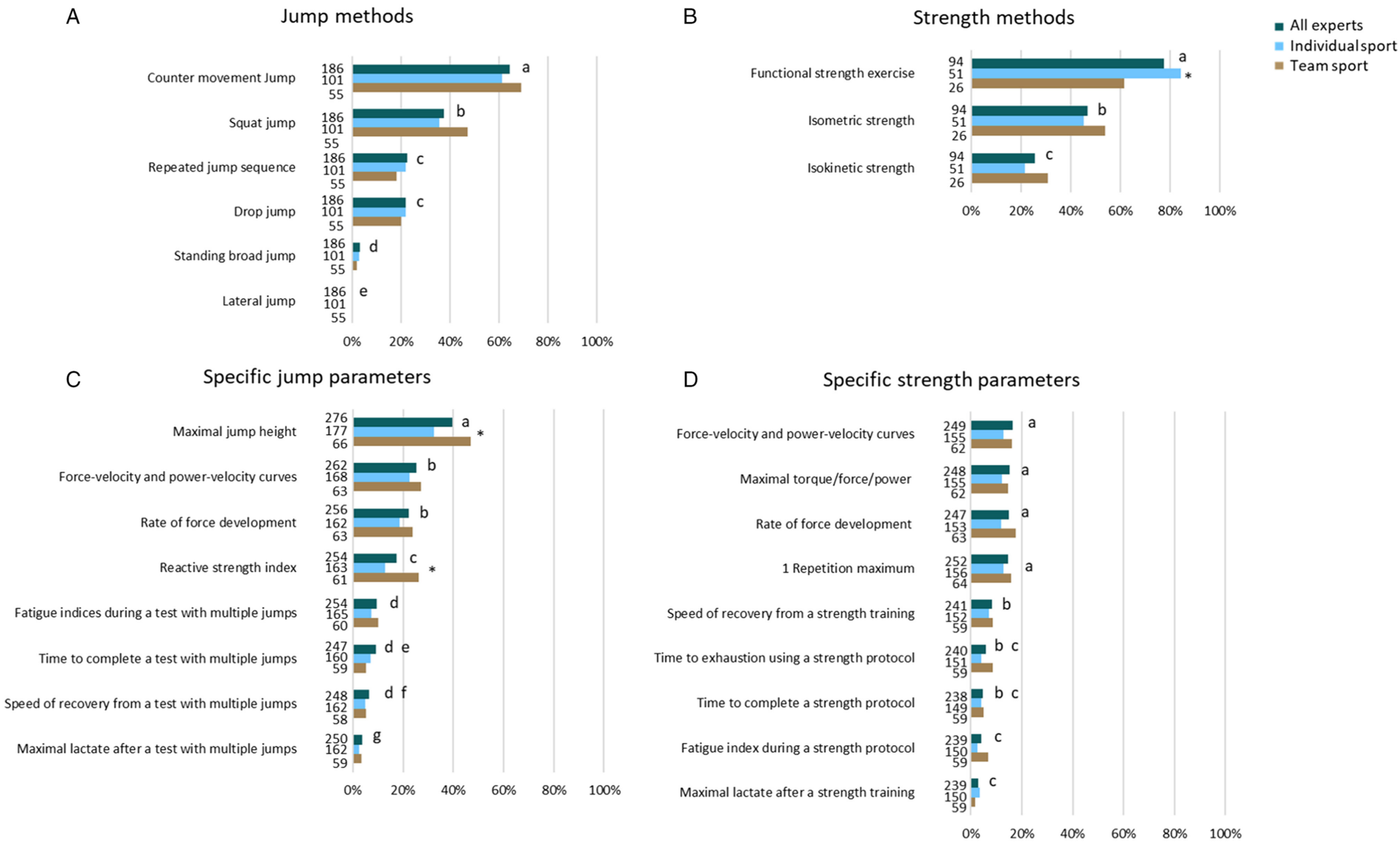

Figure 3 — Different types of jump and strength tests (A–B) and their specific components used by experts to estimate muscle fiber-type composition (C–D). The number of experts who answered each question can be seen on the left side of the bar chart. *Significant differences between individual- and team-sport experts. Identical letters indicate that there is no significant difference in the percentage of use of different methods in the total group of experts.

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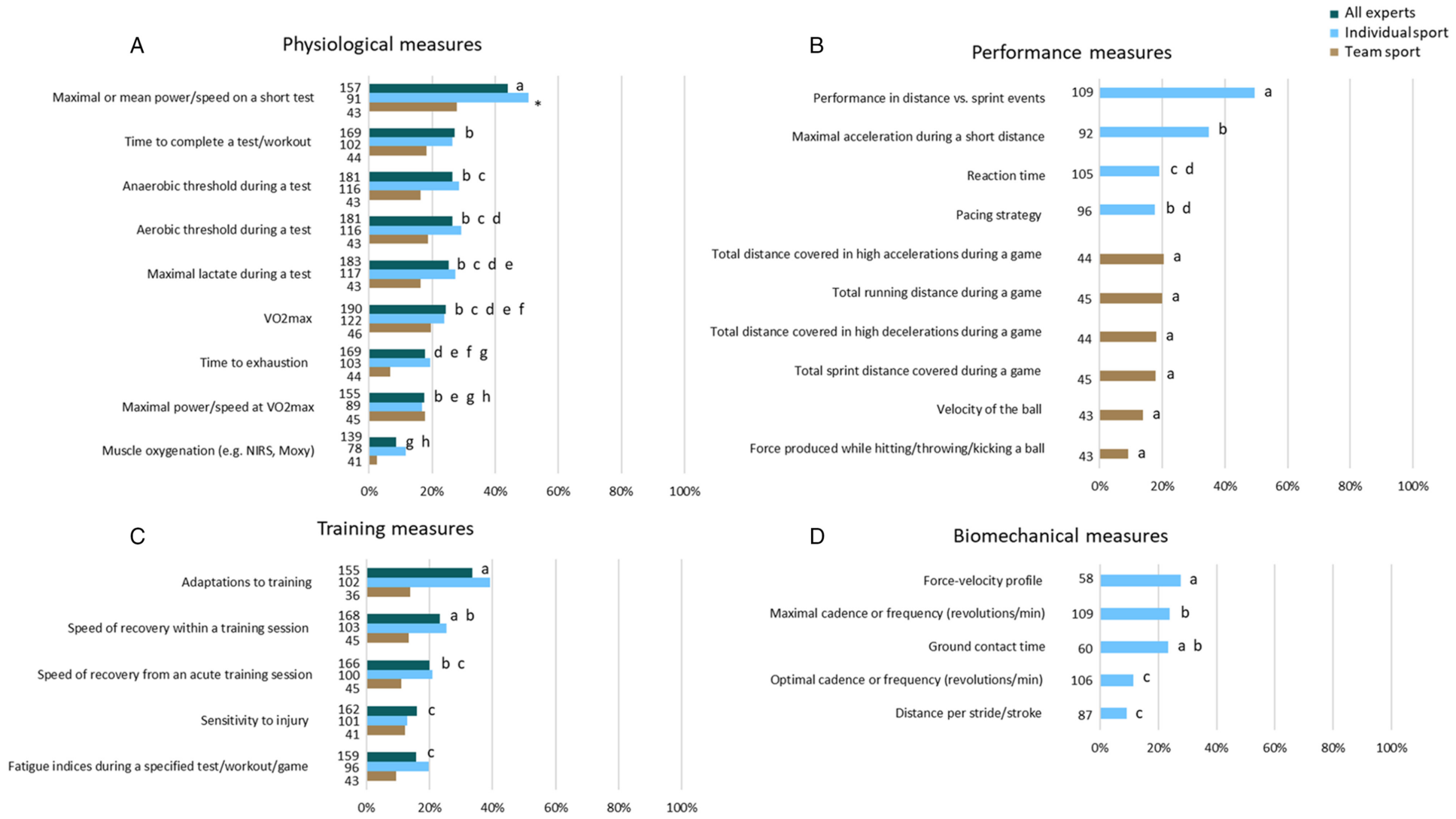

Figure 4 — Experts' use of sport-specific methods to estimate the muscle fiber-type composition in their athletes. (B–D) Some performance measures, as well as the biomechanical measures, were only investigated in either individual- or team-sport experts. The number of experts who answered each question can be seen on the left side of the bar chart. *Significant differences between individual- and team-sport experts. Identical letters indicate that there is no significant difference in the use of different methods in the total group of experts.

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mean power/speed on a short test (43.9%) was most frequently used to estimate MFTC. Moreover, this test was more frequently used by individual sport experts when compared to team sport experts (50.5% vs 27.9%, $P = .014$).

When looking into training measures that were interpreted to be related to MFTC, adaptations to training (33.5%) and speed of recovery within an acute training session (23.2%) were more often used to get an impression of the MFTC than speed of recovery from an acute training session (19.9%), sensitivity to injury (16.0%), and fatigue indices during a specific test/workout/game (15.7%).

Performance measures were queried sport-specifically. In the individual sports, almost half of the experts used performance in distance versus sprint events to estimate MFTC (Figure 4B). In team sports, performance measures such as total distance covered in high accelerations and total running distance were used by 20.5% to 9.30% of the experts to estimate MFTC.

Lastly, in individual sport experts, the use of some biomechanical test parameters to estimate MFTC was inquired. The force–velocity profile (27.6%) was mostly used to estimate MFTC.

All above-discussed sport-specific methods were not different when comparing experts' roles nor level of the experts' athletes in the team or individual sport experts, respectively.

What MFTC Is Used for During Training and Competitions

Experts who were currently using MFTC during their training and competition were asked for which decisions they used this knowledge (Figure 5A). Interestingly more than 80% of these experts used this knowledge to individualize training volume/duration and intensity. Moreover, approximately 50% to 60% of the experts used this information to individualize duration of recovery, training frequency, and taper strategy, to decide on tactical issues (pacing et cetera), and to identify talent. Lastly, one-fifth of the experts used the knowledge on MFTC to individualize nutrition. Interestingly, the information on MFTC was used to a higher extent by individual sport experts than by team sport experts to individualize taper strategy as well as to decide on tactical issues (respectively, 58.3% vs 29.4%; $P = .032$ and 63.9% vs 17.6%; $P < .001$).

Moreover, we also inquired experts on the importance of MFTC for coaching practice, irrespective of whether they currently used MFTC in their training practice or not (Figure 5B). On average, experts rated the importance of MFTC from 3.23 to 4.03 (moderately important to important) for all decisions. Individualizing tactical issues based on MFTC was shown to be more important for individual sport experts ($P < .001$), while the importance of MFTC for individualization of nutrition was perceived more important by team sport experts ($P = .048$).

The use and importance of the MFTC were not different when comparing expert's roles nor level of the experts' athletes in the team/individual sport experts, respectively.

Discussion

The present study surveyed 446 experts to investigate current beliefs on the practical importance of MFTC in sport and the used noninvasive versus invasive methods to estimate MFTC. Because of the wide network of our research team, a large number of high-standard and highly educated experts were reached. Experts consisted of both sport-specific coaches and SSSM staff. In general, almost no differences in opinions were found between both groups when analyzed for each sport group separately. Only a minority of the respondents were female (11%), which is a significant gender bias. But, unfortunately, this percentage is representative for contemporary high-standard coaching staffs as the Olympic Committee recently published that only 13% of the coaches at the 2020 Tokyo Olympics were female.[27]

In total, 90% of the experts expressed an intention to use the knowledge on MFTC in their coaching practice. This clearly indicates that experts recognize the importance of MFTC in guiding their athletes. However, 50% of these experts could not currently use this knowledge since they are unable to estimate MFTC. This is probably partly related to the current gold standard for measuring MFTC, which is via an invasive muscle biopsy, as less than 20% of all experts believe their athletes would be willing to undergo this procedure. Team sport experts perceive this willingness to be even lower, which might be related to the fact that research in the 70s mostly focused on the MFTC of individual athletes, showing relevance for talent identification, while MFTC was less investigated in team sports.[5–8] However, more recently, the importance of MFTC was also shown for team sports, as it was found to be related to the decline in high-intensity running activities during a soccer game, as well as to hamstring strain injuries in elite soccer players.[25,28,29] Also, it could be hypothesized that taking biopsies is less easy to organize in a group of team sport athletes when compared to individual athletes. Despite the reluctance to use biopsies, 40% of the experts are completely or fairly confident in ranking their athletes on a continuum from fast to slow, indicating that other noninvasive estimators are used to gain insights into MFTC. Although coaches appear to be confident in the validity of their assessments, it is questionable whether all used methods are valid for estimating MFTC.

In general, experts rely heavily on subjective methods, like experience and visual inspection, to estimate MFTC. This is especially true for individual experts, who are relying mostly on experience while team sport experts depend more on jump tests. This may be due to the fact that individual coaches typically work with their athletes for longer periods. When an expert trains an athlete for multiple years, subjective indicators might be a valid way of estimating the MFTC. However, it is far more difficult, and arguably more important, to estimate MFTC if an athlete is unknown to the expert, which might occur more frequently in team sport. In this way, objective noninvasive measurements like jump, strength, and power tests could be used to estimate MFTC. Although jump tests are often performed in the general screening of team sport athletes, the questionnaire explicitly stated that coaches had to report methods that informed them about the MFTC, not any method that they routinely use.

The validity of using experience and visual inspection depends on the parameters on which these observations are based. The scientific evidence is strong for fatigue during training/competition[20,30] but is not for morphology[31] and sprint times.[14] For example, visual inspection is used by almost 25% of the experts, despite the fact that the limited available literature shows no correlation between MFTC and muscle volume in untrained individuals.[31] Nonetheless, the fact that most experts associate muscle size with MFTC is probably due to the observation that athletes in sports where a fast MFTC is prevalent, such as sprint disciplines, also require high muscle mass. This does not mean, however, that muscle volume is directly related to the MFTC itself in all populations.

Various other noninvasive tests have shown to be related to MFTC. However, results are conflicting (depending on

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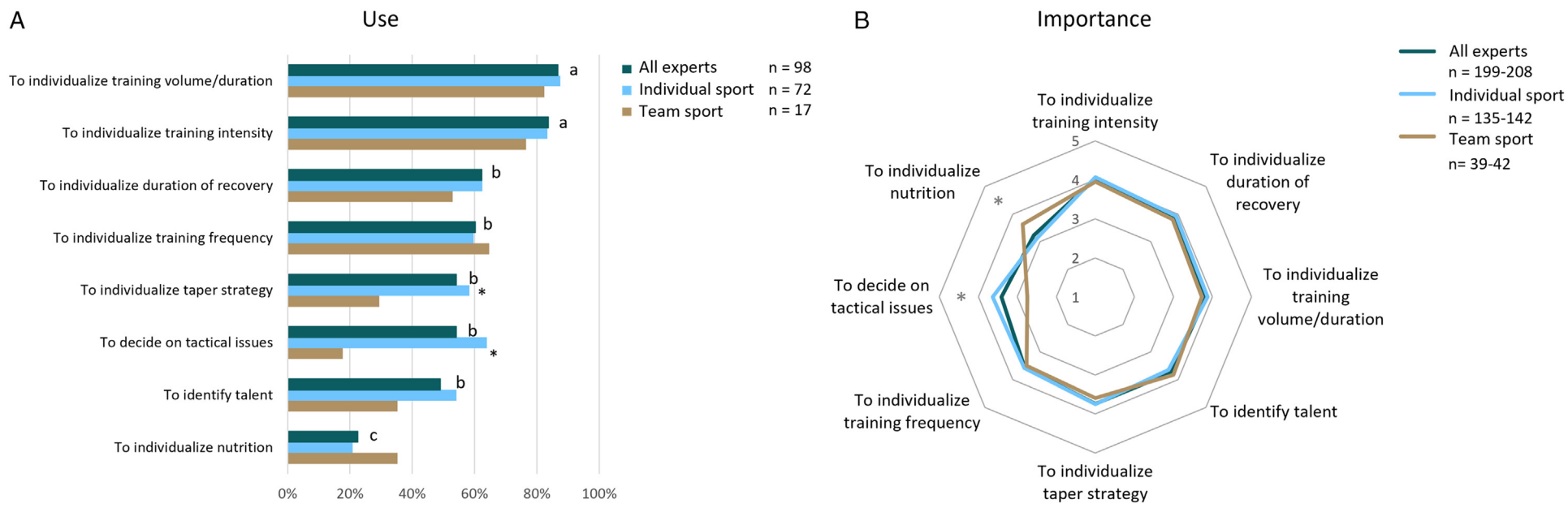

Figure 5 — The use (A) and the perceived importance (B) of the muscle fiber-type composition for different coaching decisions. Number of experts who answered the question is shown in the figure. *Significant differences between individual- and team-sport experts. Identical letters indicate that there is no significant difference in the use of different methods in the total group of experts. (1 = not important, 2 = slightly important, 3 = moderately important, 4 = important, and 5 = very important).

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participants, sample size, study design, correlation to either type I/type IIa/type IIx fibers, and correlation to percentage of number or percentage of area of the fibers, etc), and correlations are often moderate, indicating that most tests are not able to adequately estimate MFTC. Performance during single[18,19] and repeated jump[12] tests have been shown to be related to MFTC. In countermovement jumps, a higher jump height is being associated with a higher proportion of fast fibers. However, the link between strength tests and MFTC is less straightforward. The most frequently used parameters to estimate MFTC in strength tests were either maximal force or velocity-related parameters (maximal power, force–velocity curves, and rate of force development). Nevertheless, the current literature is unclear on whether individuals with a faster MFTC are stronger and show a higher maximal force compared to those with a slower MFTC. Conversely, force–velocity properties have been repeatedly linked to the MFTC and seems more appropriate when assessing MFTC.[32,33] Also, power during cycling exercises is often used as an indicator of MFTC, although correlations between the relative peak power during a Wingate test and the proportion of fast fibers may be small.[14]

Experts can generally recognize fast and explosive players/athletes, but this does not necessarily represent an individual athletes' MFTC. It is important to note that being fast is mostly assessed using methods experts have at hand or that are traditionally available in their sport. Team sport experts mainly look at GPS-derived parameters, like mean or maximal speed or maximal height at jump tests, while individual sport experts look at performance in long versus short events. Additionally, being fast not only depends on the interpreted parameter but also on multiple factors next to the MFTC like technique, muscle volume, reaction time, and anticipation.[34,35] When using sprint tests, weak correlations are usually observed between performance and MFTC.[14,18] Inbar et al[14] found no relationship between the running speed during a 40-m sprint test and the proportion of fast-twitch fibers ($r = .34$), while Korhonen et al[18] found a significant negative relationship between a 60-m running time and the % abundance of fast-twitch fibers ($r = -.26$) in a group of 91 sprint-trained athletes (18–84 y). Also, despite the fact that almost 1 out of 5 individual sport experts use reaction time to interpret MFTC, there is only one study in children linking reaction time to MFTC.[36] Clearly, there is a discrepancy between the scientific literature and sports practice. However, it is unclear whether information is insufficiently communicated to experts, or whether research is lagging behind, or both.

Experts are using the MFTC for important decisions during training/competition. More than 80% of the experts utilize MTFC to individualize training volume and intensity. It has been shown that athletes with a slower MFTC handle a higher training volume better than subjects with a faster MFTC.[23] Moreover, according to the Henneman's size principle, intensity is important to recruit fast-twitch fibers, which is essential for these fibers to adapt.[37] Therefore, individuals with a different MFTC might require individualized training, in which individuals with a fast MFTC need more short efforts at critical speed and repeated sprint, while individuals with a slow MFTC are in need of longer slow continuous efforts, as well as some longer efforts at critical speed.[38] Surprisingly, MFTC information is employed to a greater extent for individualizing training than for identifying talent (49.0%). This finding is unexpected as several studies describe its value for talent identification.[4,5,39] Furthermore, 20% of experts also report the use of MFTC to individualize nutrition, even though there is no direct evidence available on this topic. It seems possible that, in this area, the research community is lagging behind the field. Therefore, experts' experiences could be seen as very valuable insights, as a good source of inspiration and as a suggestion to further set the research agenda.

It should be noted that muscle biopsies (13.2%), as well as techniques like proton magnetic resonance spectroscopy (11.8%), are shown to be used by a noteworthy number of experts. This might indicate that some of the respondents were closely involved with academia, which corresponds to the high percentage of experts with a high educational standard. It could be questioned whether these results can be generalized to the entire coaching population. However, the increasing professionalization and growing financial resources in competition sports nowadays attract highly educated experts, becoming a new standard. Despite a predominance of trained and experienced experts, 7% of the respondents did not complete any training regarding physiology of exercise and can therefore not be classified as real experts.

Coaching practices regarding the implementation of MFTC knowledge in training and competition precede scientific evidence. Further research is necessary to validate or challenge current practices. However, to conduct adequately powered studies in athletes, noninvasive techniques for estimating MFTC are required. Currently, it remains unclear which noninvasive technique is best suited for predicting MFTC, as the majority of the existing research has only examined the correlation between biopsies and one single noninvasive technique. A direct comparison of multiple noninvasive techniques against the gold standard is thus needed. Additionally, the accuracy, stability, usability, and cost of potential noninvasive estimators should be assessed to ensure they are suitable for the current coaching practice.

Practical Applications

Although the concept of MFTC is taught in every exercise physiology course, it remained unclear whether this information was actually applied in sports practice. This survey uncovered that the concept of MFTC is extensively used by high-performance experts, who consider this information highly relevant for many different aspects of coaching practice. Since the majority of experts are reluctant to use muscle biopsies, numerous alternative noninvasive estimators of MFTC are currently employed. However, not all techniques and parameters are supported by scientific evidence, partly because the research is not available and partly because the negative findings of available research have not been passed on to coaching practice. The latter offers opportunities for optimizing experts' education on this topic.

Conclusions

Ninety percent of the experts either use muscle fiber-type composition (MFTC) in their coaching practice or express a desire to implement it in future. In order to estimate MFTC, experts rely on numerous noninvasive methods. However, some of these estimators lack robust scientific support, suggesting a need for education and further research in this area. Experts value MFTC for their coaching practice and use it in particular to tailor training to the individual.

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